Saba Dadsetan

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LinkedIn | GitHub | Google Scholar | Personal Website

Summary

Expert in Computer Vision, Machine Learning, and Deep Learning with a track record of impactful research and industry application in advanced medical imaging and AI-driven solutions. Experienced in deploying models for real-world applications, including Alzheimer's disease progression, cancer risk prediction, and aerial imagery analysis.

Experience Highlights

Abbvie, San Francisco, CA
Senior AI Scientist May 2025 - Present

- Working on Foundation Models for Precision Medicine, Digital Pathology team.

Genentech, San Francisco, CA
AI Research Engineer Sep 2022 - May 2024

- Modeled a new end-point for Alzheimer's Disease progression using advanced machine learning techniques.
- Developed vision language models (VLM) for prognostic tasks, enhancing predictive accuracy.

AI Research Intern Jun 2022 - Sep 2022

- Engineered a self-supervised framework for prognostic tasks; published in TMLR.

University of Pittsburgh, Pittsburgh, PA
Graduate Research Assistant Aug 2018 - Aug 2024

- Analyzed scanner effects on Alzheimer's progression prediction using multimodal medical imaging.
- Developed models for the prediction of breast cancer risk and the classification of pancreatic cystic lesions.
- Built machine learning models for osteoporosis risk prediction using x-ray scans.
- Created a semi-supervised 3D brain tumor segmentation method leveraging 2D medical images.

IntelinAir, Champaign, IL
Graduate Research Intern May 2020 - Aug 2020

- Implemented prediction and detection models for longitudinal aerial imagery, published in AAAI.
- Built a graph convolution-based model for segmentation, published in CVPRW.
- co-Led a session of WiML virtual workshop at ICML 2020.

Education

University of Pittsburgh
Ph.D in Computer Science 2017 - 2024

- Dissertation: Enhancing Alzheimer's prognostic models with cross-domain self-supervised learning and MRI data harmonization.
- Advisor: Prof. Dana L. Tudorascu
- Relevant Coursework: Artificial Intelligence, Advanced Machine Learning, Computer Vision, Algorithm Design, Game Design (Carnegie Mellon University)

University of Tehran
B.Sc. in Computer Engineering 2012 - 2017

Technical Skills

- **Programming Languages:** Python, C++, MATLAB, Java
- **Deep Learning Frameworks:** TensorFlow, PyTorch, PyTorch Lightning
- **Computer Vision Techniques:** 3D reconstruction, segmentation, image classification, object detection
- **Tools & Platforms:** Git, AWS (S3, EC2), Docker, Kubernetes

- **Web Development:** HTML, CSS, JS

Patents

- **S. Dadsetan**, G. Rose, N. Hovakimyan and J. Hobbs, Intelinair Inc., 2022. Nutrient Deficiency Detection and Forecasting System. U.S. Patent Application 17/676,474.

Publications

1. Robust Alzheimer's Progression Modeling using Cross-Domain Self-Supervised Deep Learning.
   S. Dadsetan, M. Hejrati, S. Wu, S. Hashemifar. Medical Transactions on Machine Learning Research (TMLR) 2023: https://openreview.net/forum?id=HVAeM6sNo8.
2. Anterior cruciate ligament classification using pseudo-masking and hard attention.
   S. Dadsetan, M. Albers, A. Weinstock, V. Musahl, D. Arefan, S. Wu. Medical Imaging 2023: Computer-Aided Diagnosis.
3. Deep learning of longitudinal mammogram examinations for breast cancer risk prediction.
   S. Dadsetan, D. Arefan, W. Berg, M. L. Zuley, J. H. Sumkin, S. Wu. Pattern Recognition 132 (2022): 108919.
4. No-Reference Image Quality Assessment via Transformers, Relative Ranking, and Self-Consistency
   S. A. Golestaneh, **S. Dadsetan**, K. M Kitani - WACV 2021 (Oral)
5. Radiomics-informed Deep Curriculum Learning for Breast Cancer Diagnosis
   G. Nebbia , **S. Dadsetan**, D.Arefan, M.L. Zuley, J.H. Sumkin, H. Huang, S. Wu. Proceedings of the International Conference on Medical image computing and computer-assisted intervention – MICCAI 2021.
6. Superpixels and Graph Convolutional Neural Networks for Efficient Detection of Nutrient Deficiency Stress from Aerial Imagery
   S. Dadsetan, D. Pichler, D. Wilson, N. Hovakimyan, J. Hobbs. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops – CVPRW 2021.
7. Detection and Prediction of Nutrient Deficiency Stress using Longitudinal Aerial Imagery
   S. Dadsetan,G. Rose, N. Hovakimyan, J. Hobbs. Proceedings of the Thirty-fifth AAAI conference on artificial intelligence – AAAI 2021.
8. Learning Knowledge from Longitudinal Data of Mammograms to Improving Breast Cancer Risk Prediction
   S. Dadsetan and S. Wu. Medical Imaging 2021: Imaging Informatics for Healthcare, Research, and Applications.
9. A data interpretation approach for deep learning-based prediction models.
   S. Dadsetan and S. Wu. Medical Imaging 2019: Imaging Informatics for Healthcare, Research, and Applications, vol. 10954, p. 109540M

Honors and Awards

- Google GHC Scholarship Recipient, 2021
- 3rd place Award for ACM Student Research Competition at Tapia Conference, 2020
- ACM Richard Tapia Conference Student Scholarship, 2020
- Grace Hopper Celebration Student Scholarship, 2019
- Fellowships from the Intelligent Systems Program, 2017-2018
- Iranian National Olympiads in Mathematics & Informatics, Semi-finalist, 2010-2011